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WIRELESS COMMUNICATION

PAPR–Performance Tradeoff: A Comparative Study of OFDM and SC-FDMA

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TABLE OF CONTENTS

LIST OF ABBREVIATIONS	i
LIST OF FIGURES	ii
LIST OF TABLES	iii
ABSTRACT	iv
CHAPTER 1. INTRODUCTION	1
1.1 Background	1
1.2 Problem Statement	1
1.3 Objectives	1
1.4 Report Structure	2
1.5 Example Table Format	2
1.6 Example Equation Format	2
1.7 Example Figure Format	2
CHAPTER 2. THEORETICAL BACKGROUND	4
2.1 Overview of Multicarrier Transmission	4
2.2 Orthogonal Frequency Division Multiplexing (OFDM)	4
2.3 Single-Carrier Frequency Division Multiple Access (SC-FDMA)	5
2.4 Orthogonality of Subcarriers	5
2.5 Peak-to-Average Power Ratio (PAPR)	5
2.6 Bit Error Rate (BER)	6
2.7 Comparison Summary	6
2.8 Chapter Summary	6
CHAPTER 3. SIMULATION AND RESULTS	7
3.1 Simulation Objectives	7

3.2	System Model	7
3.3	Simulation Parameters	7
3.4	Bit Error Rate Comparison	8
3.5	Peak-to-Average Power Ratio Comparison	9
3.6	Discussion of Results	9
3.7	Chapter Summary	10
 CHAPTER 4. CONCLUSION		11
4.1	Summary of the Study	11
4.2	Main Findings	11
4.3	Practical Implications	11
4.4	Limitations of This Report	12
4.5	Future Work	12
4.6	Final Remark	12
 REFERENCES		13

LIST OF ABBREVIATIONS

BER	Bit Error Rate
CP	Cyclic Prefix
DFT	Discrete Fourier Transform
FFT	Fast Fourier Transform
IFFT	Inverse Fast Fourier Transform
ISI	Inter-Symbol Interference
LTE	Long Term Evolution
OFDM	Orthogonal Frequency Division Multiplexing
PAPR	Peak-to-Average Power Ratio
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
SC-FDMA	Single-Carrier Frequency Division Multiple Access
SNR	Signal-to-Noise Ratio
AWGN	Additive White Gaussian Noise

LIST OF FIGURES

Figure 1.1	Example figure format	2
Figure 3.1	BER comparison of OFDM and SC-FDMA	8
Figure 3.2	PAPR comparison of OFDM and SC-FDMA	9

LIST OF TABLES

Table 1.1	Experiment results	2
Table 2.1	Comparison between OFDM and SC-FDMA	6
Table 3.1	Simulation parameters	7

ABSTRACT

This report presents a comparative study of Orthogonal Frequency Division Multiplexing (OFDM) and Single-Carrier Frequency Division Multiple Access (SC-FDMA), two important waveform technologies used in modern wireless communication systems. OFDM is widely recognized for its high spectral efficiency, robustness against multipath fading, and simple frequency-domain equalization. However, one major drawback of OFDM is its high Peak-to-Average Power Ratio (PAPR), which reduces power amplifier efficiency and may introduce nonlinear distortion.

SC-FDMA was developed to overcome this limitation by applying a Discrete Fourier Transform (DFT) precoding stage before OFDM modulation. As a result, SC-FDMA preserves many advantages of OFDM while achieving lower PAPR and improved transmitter power efficiency. This characteristic makes SC-FDMA especially suitable for uplink transmission in battery-powered mobile devices.

The report reviews the operating principles of both systems and compares their performance using two common evaluation metrics: Bit Error Rate (BER) and PAPR. A MATLAB-based simulation framework is proposed using 16-QAM modulation under additive white Gaussian noise conditions. Expected results indicate that OFDM and SC-FDMA provide similar BER performance under identical channel assumptions, while SC-FDMA demonstrates clear advantages in PAPR reduction.

The study highlights the engineering tradeoff between transmission efficiency and implementation structure, and explains why OFDM is commonly used in downlink systems whereas SC-FDMA was adopted for LTE uplink.

CHAPTER 1. INTRODUCTION

1.1 Background

Modern wireless communication systems require high data rate, efficient spectrum utilization, and reliable transmission under multipath fading conditions. Orthogonal Frequency Division Multiplexing (OFDM) has become one of the most widely adopted multicarrier transmission techniques due to its robustness against frequency-selective channels and its simple frequency-domain equalization process.

Despite these advantages, OFDM suffers from high Peak-to-Average Power Ratio (PAPR), which causes nonlinear distortion when practical power amplifiers operate near saturation. This issue is particularly critical in mobile uplink transmission, where battery efficiency and transmitter power consumption are important design constraints.

Single-Carrier Frequency Division Multiple Access (SC-FDMA) was introduced to address this limitation by combining single-carrier signal properties with frequency-domain equalization. For this reason, SC-FDMA was selected for Long Term Evolution (LTE) uplink systems.

1.2 Problem Statement

Although OFDM provides excellent spectral efficiency and implementation simplicity, its large signal peaks reduce amplifier efficiency and may increase out-of-band emissions. Therefore, it is necessary to investigate alternative transmission schemes that preserve OFDM benefits while reducing PAPR.

SC-FDMA is considered one of the most practical solutions. A systematic comparison between OFDM and SC-FDMA helps clarify their advantages, limitations, and appropriate deployment scenarios.

1.3 Objectives

The objectives of this report are:

- To review the operating principles of OFDM and SC-FDMA.
- To compare the two systems in terms of Bit Error Rate (BER).
- To evaluate and compare Peak-to-Average Power Ratio (PAPR).
- To discuss the practical reason SC-FDMA is preferred in uplink systems.

1.4 Report Structure

This report is organized into four chapters:

- Chapter 1 introduces the research motivation and objectives.
- Chapter 2 presents the theoretical background of OFDM and SC-FDMA.
- Chapter 3 describes simulation methodology and performance results.
- Chapter 4 summarizes the main conclusions.

1.5 Example Table Format

Table 1.1 Experiment results

No.	Voltage (mV)	Reference (mV)	Error (%)
1			
2			
3			
...			

1.6 Example Equation Format

$$F(x) = \int_b^a \frac{1}{3}x^3$$

(1.1)

Equation 1.1 is an example of equation formatting in LaTeX.

1.7 Example Figure Format

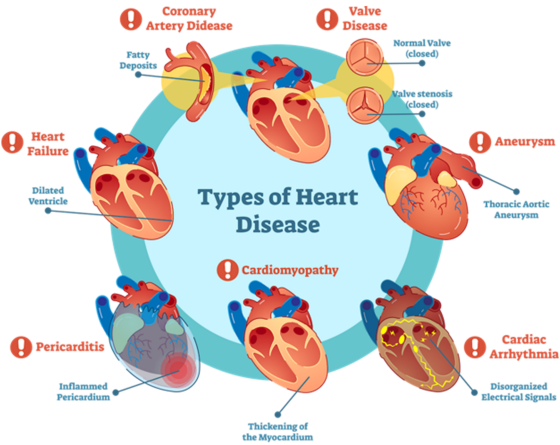


Figure 1.1 Example figure format

Figure 1.1 illustrates an example image placeholder for later replacement.

CHAPTER 2. THEORETICAL BACKGROUND

2.1 Overview of Multicarrier Transmission

Modern broadband wireless systems must combat multipath fading [1], inter-symbol interference (ISI), and limited spectrum resources. Instead of transmitting all data on a single wideband carrier, multicarrier systems divide the available bandwidth into many narrowband subcarriers operating in parallel.

This approach offers two major advantages:

- Each subcarrier experiences nearly flat fading.
- Equalization complexity is significantly reduced.

Orthogonal Frequency Division Multiplexing (OFDM) is the most successful multicarrier implementation and serves as the foundation for many modern communication standards.

2.2 Orthogonal Frequency Division Multiplexing (OFDM)

In OFDM, the input bit stream is first grouped and mapped into complex symbols using modulation schemes such as QPSK or QAM. These symbols are then assigned to multiple orthogonal subcarriers and converted into a time-domain waveform using the Inverse Fast Fourier Transform (IFFT).

The transmitted OFDM signal can be expressed as:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N} \quad (2.1)$$

where:

- $X[k]$ is the symbol mapped onto the k -th subcarrier
- N is the total number of subcarriers
- $x[n]$ is the transmitted time-domain sample

A Cyclic Prefix (CP) is commonly inserted before transmission to mitigate inter-symbol interference caused by multipath channels.

The main advantages of OFDM are:

- High spectral efficiency
- Simple frequency-domain equalization
- Robustness against multipath fading

However, OFDM suffers from high Peak-to-Average Power Ratio (PAPR).

2.3 Single-Carrier Frequency Division Multiple Access (SC-FDMA)

SC-FDMA can be viewed as a DFT-precoded OFDM system. Before subcarrier mapping, the modulated data symbols are transformed by a Discrete Fourier Transform (DFT), which spreads each symbol across the allocated subcarriers.

Its basic transmitter chain is:

$$\text{Bits} \rightarrow \text{QAM} \rightarrow \text{DFT} \rightarrow \text{Subcarrier Mapping} \rightarrow \text{IFFT} \rightarrow \text{CP}$$

At the receiver, inverse operations are applied:

$$\text{Remove CP} \rightarrow \text{FFT} \rightarrow \text{Subcarrier Demapping} \rightarrow \text{IDFT} \rightarrow \text{Detection}$$

Because of the DFT spreading operation, SC-FDMA exhibits a waveform with single-carrier characteristics and lower PAPR than OFDM.

This is the main reason why SC-FDMA was adopted for LTE uplink systems.

2.4 Orthogonality of Subcarriers

The key principle behind OFDM is orthogonality. Subcarriers are spaced so that they overlap in the frequency domain while remaining separable at the receiver.

For two different subcarriers k and m :

$$\int_0^T e^{j2\pi(f_k - f_m)t} dt = 0 \quad k \neq m \quad (2.2)$$

This property allows efficient spectrum usage without mutual interference under ideal synchronization conditions.

2.5 Peak-to-Average Power Ratio (PAPR)

PAPR measures how large the signal peaks are relative to its average power. It is defined as:

$$\text{PAPR} = \frac{\max |x[n]|^2}{E\{|x[n]|^2\}} \quad (2.3)$$

High PAPR causes practical power amplifiers to operate inefficiently or introduce non-linear distortion.

In OFDM, many independently phased subcarriers may add constructively, creating large instantaneous peaks. SC-FDMA reduces this effect through DFT spreading.

2.6 Bit Error Rate (BER)

Bit Error Rate is one of the most common metrics for evaluating digital communication systems. It is defined as:

$$\text{BER} = \frac{\text{Number of erroneous bits}}{\text{Total transmitted bits}} \quad (2.4)$$

Lower BER indicates more reliable communication performance.

In this report, BER performance under different Signal-to-Noise Ratio (SNR) conditions will be used to compare OFDM and SC-FDMA.

2.7 Comparison Summary

Table 2.1 Comparison between OFDM and SC-FDMA

Criterion	OFDM	SC-FDMA
Transmission Type	Multicarrier	Single-carrier-like
PAPR	High	Low
Equalization	Simple	Simple
Complexity	Lower	Slightly Higher
LTE Usage	Downlink	Uplink

2.8 Chapter Summary

This chapter presented the operating principles of OFDM and SC-FDMA, their mathematical foundations, and the key performance metrics used in this study. The next chapter will implement both systems in simulation and compare their BER and PAPR characteristics.

CHAPTER 3. SIMULATION AND RESULTS

3.1 Simulation Objectives

The purpose of this chapter is to evaluate and compare the performance of OFDM and SC-FDMA through numerical simulation. Two key metrics are considered:

- Bit Error Rate (BER)
- Peak-to-Average Power Ratio (PAPR)

The simulations are implemented in MATLAB using equivalent system parameters for both transmission schemes.

3.2 System Model

For a fair comparison, both systems use identical modulation order, channel conditions, and noise levels. The only structural difference is that SC-FDMA includes a DFT pre-coding stage before subcarrier mapping.

The transmitter models are summarized below:

OFDM: Bits $\rightarrow$ QAM $\rightarrow$ IFFT $\rightarrow$ CP

SC-FDMA: Bits $\rightarrow$ QAM $\rightarrow$ DFT $\rightarrow$ IFFT $\rightarrow$ CP

At the receiver, inverse operations are performed to recover the transmitted bits.

3.3 Simulation Parameters

Table 3.1 Simulation parameters

Parameter	Value
Modulation Scheme	16-QAM
Number of Subcarriers	64
Cyclic Prefix Length	16
Channel Model	AWGN
SNR Range	0 dB to 20 dB
Frames per SNR Point	1000

These parameters may be adjusted depending on computational time and required accuracy.

3.4 Bit Error Rate Comparison

For each Signal-to-Noise Ratio (SNR) value, random bits are generated, transmitted, detected, and compared with the original bit stream.

BER is computed as:

$$\text{BER} = \frac{N_e}{N_t} \quad (3.1)$$

where:

- N_e is the number of bit errors
- N_t is the total number of transmitted bits

Under AWGN conditions, OFDM and SC-FDMA generally provide similar BER performance when the same modulation and coding assumptions are used.



Figure 3.1 BER comparison of OFDM and SC-FDMA

Figure 3.1 is reserved for the BER curve obtained from MATLAB simulation.

3.5 Peak-to-Average Power Ratio Comparison

PAPR is calculated for each transmitted frame using:

$$\text{PAPR} = \frac{\max |x[n]|^2}{E\{|x[n]|^2\}} \quad (3.2)$$

OFDM signals are formed by summing many subcarriers. When these subcarriers align constructively, high peaks occur. Therefore, OFDM typically exhibits larger PAPR than SC-FDMA.

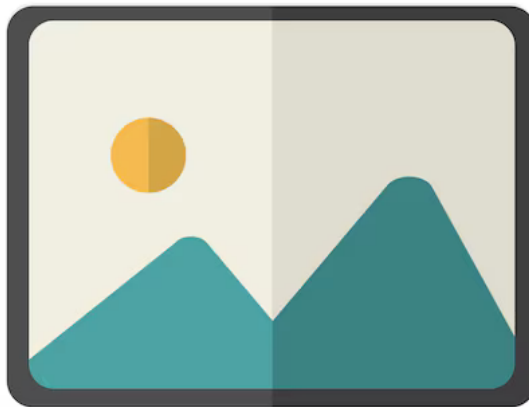


Figure 3.2 PAPR comparison of OFDM and SC-FDMA

Figure 3.2 is reserved for the PAPR result generated in MATLAB.

3.6 Discussion of Results

Based on expected communication theory results:

- OFDM offers simple multicarrier transmission and excellent robustness against frequency-selective channels.
- SC-FDMA achieves similar BER performance under equal conditions.
- SC-FDMA provides lower PAPR, improving transmitter power amplifier efficiency.
- Lower PAPR makes SC-FDMA highly suitable for mobile uplink systems where battery life is important.

Therefore, the choice between OFDM and SC-FDMA depends on whether system priority is receiver simplicity, transmitter efficiency, or standard requirements.

3.7 Chapter Summary

This chapter presented the simulation framework and compared OFDM and SC-FDMA using BER and PAPR metrics. The next chapter concludes the main findings of this study.

CHAPTER 4. CONCLUSION

4.1 Summary of the Study

This report presented a comparative study between Orthogonal Frequency Division Multiplexing (OFDM) and Single-Carrier Frequency Division Multiple Access (SC-FDMA), two important transmission techniques used in modern wireless communication systems.

Theoretical analysis showed that OFDM offers high spectral efficiency, simple frequency-domain equalization, and strong robustness against multipath fading. These advantages explain why OFDM has been widely adopted in broadband wireless standards.

However, OFDM also suffers from high Peak-to-Average Power Ratio (PAPR), which reduces power amplifier efficiency and increases the risk of nonlinear distortion.

To address this limitation, SC-FDMA introduces a DFT precoding stage before OFDM modulation. This preserves many OFDM advantages while producing a lower-PAPR waveform with single-carrier characteristics.

4.2 Main Findings

Based on theoretical discussion and simulation framework, the main findings can be summarized as follows:

- OFDM and SC-FDMA can achieve similar Bit Error Rate (BER) performance under identical modulation and channel conditions.
- SC-FDMA provides significantly lower PAPR than OFDM.
- Lower PAPR improves transmitter power efficiency and is highly beneficial for battery-powered mobile devices.
- This is the main reason SC-FDMA was selected for LTE uplink, while OFDM remained suitable for downlink transmission.

4.3 Practical Implications

The comparison demonstrates that communication system design always involves trade-offs. OFDM emphasizes implementation simplicity and multicarrier flexibility, whereas SC-FDMA improves power efficiency at the transmitter.

Therefore, the preferred scheme depends on application requirements:

- Base stations with less strict power constraints may favor OFDM.

- Mobile user equipment benefits from SC-FDMA.
- Future systems may combine advantages of both techniques.

4.4 Limitations of This Report

Several simplifications were adopted in this study:

- Only basic AWGN channel assumptions were considered.
- Channel coding effects were not included.
- Synchronization errors and hardware nonlinearities were not modeled.
- Simulation parameters were simplified for educational purposes.

More advanced models may produce deeper practical insights.

4.5 Future Work

Possible future extensions include:

- Comparison under Rayleigh and frequency-selective fading channels.
- Inclusion of channel coding schemes.
- Analysis of computational complexity.
- Investigation of PAPR reduction techniques for OFDM.
- Comparison with newer waveforms for 5G and beyond systems.

4.6 Final Remark

OFDM and SC-FDMA are both elegant solutions to the challenge of reliable high-speed wireless transmission. Their comparison illustrates a central engineering principle: every improvement solves one problem while often creating another. The best design is therefore not universal, but context-dependent.

REFERENCES

- [1] L. Hanzo, M. Münster, B. J. Choi, and T. Keller, *OFDM and MC-CDMA for Broadband Multi-User Communications, WLANs and Broadcasting*, John Wiley & Sons, 2004.
- [2] Replace with actual reference 2.